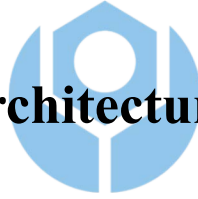


Chapter 1 System Architecture

Before delving into system details, this study first introduces the adopted system framework. To integrate the network Functional Application Platform Interface (nFAPI) with mainstream commercial equipment, the system proposed in this study adopts a stable two-stage split architecture. The network is first connected through the Split 6 nFAPI interface between network functions, and then converted to the standard O-RAN Split 7.2x interface to connect to a commercial Pegatron O-RU. Figure 1.1 illustrates this system architecture and its delay management mechanism.

1.1 End-to-End Architecture



This system establishes a complete 5G end-to-end communication link extending from the core network to the user equipment:

- **Core Network (CN):** The core component responsible for routing, session management, and user data authentication.
- **O-RAN Central Unit (O-CU):** The logical node handling higher-layer protocol stacks, such as Radio Resource Control (RRC) and Packet Data Convergence Protocol (PDCP). It connects to the distributed unit via the F1 interface.
- **O-RAN Distributed Unit (O-DU):** To optimize computational performance and deployment flexibility, the system divides the O-DU into O-DU High and O-DU Low. O-DU High executes as a Virtual

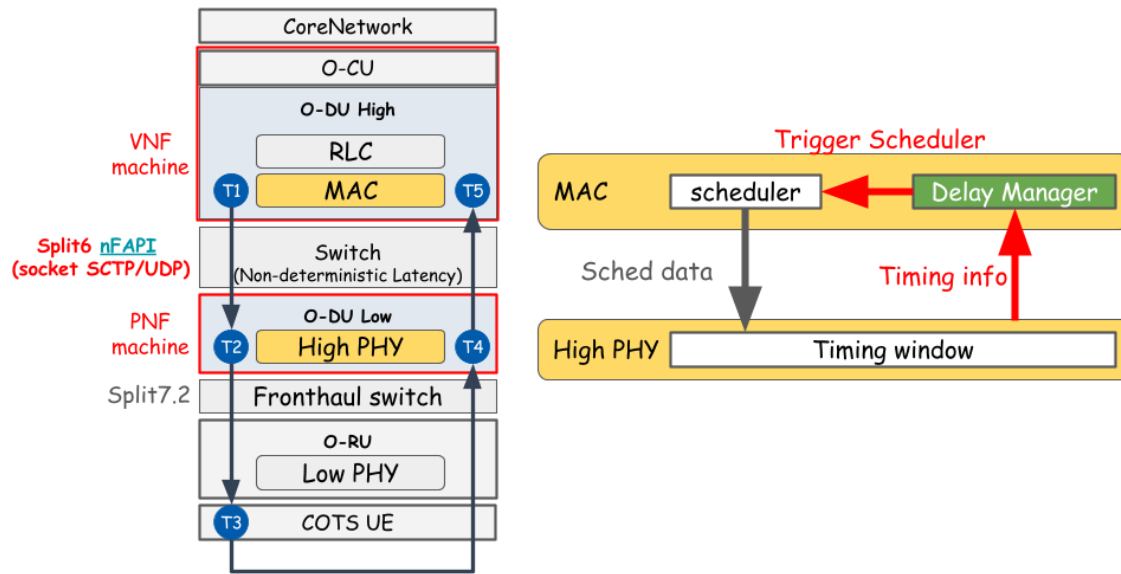


Figure 1.1: System architecture and delay management mechanism.

Network Function (VNF) on virtual machines, while O-DU Low is deployed as a Physical Network Function (PNF) on dedicated hardware.

- **Switch:** A commercial network switch located between the VNF and PNF. This intermediate layer connection utilizes the nFAPI protocol. However, traversing this packet-switched network inevitably introduces non-deterministic latency.
- **O-RAN Radio Unit (O-RU):** A dedicated hardware unit responsible for Low PHY processing, Digital Beamforming, and Radio Frequency (RF) transmission.
- **Commercial Off-The-Shelf User Equipment (COTS UE):** Standard commercial smartphones or modems acting as the network's terminal receivers.

1.2 O-DU High: MAC, Scheduler, and Timing Core

Within the O-DU High virtualized network function, the timing control is organized as a unified timing control system. The core processes handling this timing core include the Scheduler, Node sync, and Delay Management:

- **Scheduler:** Responsible for allocating available radio resources (e.g., time-frequency resource blocks) and determining the exact priority and timing of downlink/uplink data transmission.
- **Node sync:** Responsible for software-level clock synchronization. This procedure periodically exchanges synchronization timestamps ($t_1 \sim t_4$) with the PNF via standard *DL/UL_node_sync* messages. It calculates the instantaneous clock offset and applies slot-level (S_{adj}) and microsecond-level (μS_{adj}) adjustments to align the VNF's local clock to the PNF baseline, eliminating long-term hardware clock drift (Clock Drift).
- **Delay Management:** Operates on top of the stabilized clock phase established by Node sync. This mechanism utilizes the PNF Timing Info (Δt_{arrive}) feedback from the PNF to dynamically estimate transport network jitter. It then actively adjusts the slots ahead (S_{ahead}) trigger offset of the scheduler. This control compensates for the non-deterministic packet transport delays and VNF processing delays without inducing synchronization loss.

1.3 O-DU Low: High PHY and Timing Window

O-DU Low manages the High PHY baseband processing. At this level, the most critical synchronization mechanism is the nFAPI PNF Timing Window:

- **Timing Window:** A predefined and strict time interval. Let T_{window} denote the duration of this PNF timing window, and T_{slot} denote the subframe slot duration. When scheduling messages (especially `DL_TTI.request`) sent by O-DU High traverse the network, they must accurately arrive and fall exactly within this window. Otherwise, the High PHY will reject the message, resulting in the inability to process data for that slot.
- **Timing Info (Δt_{arrive}):** The PNF continuously monitors the exact timing info of control messages relative to the transmission deadline. This information is denoted as Δt_{arrive} , representing the remaining time margin within the timing window. The PNF periodically feeds this information back to the MAC layer for Delay Management. This establishes a closed-loop feedback control system to compensate for unpredictable transmission delay variations between the VNF and PNF machines.

1.4 Key Verification Metrics

To evaluate the stability, synchronization accuracy, and real-time performance of the proposed algorithm in non-ideal environments, this study defines the following key verification metrics:

- **MAC Layer HARQ RTT ($T_5 - T_1$):** Specifically measures the Hybrid Automatic Repeat Request (HARQ) loop delay at the MAC layer. This metric precisely reflects the feedback efficiency and operational health of the link layer.
- **VNF-PNF Downlink Delay ($T_2 - T_1$):** Measures the accumulated one-way downlink latency from the virtualized machine (VNF) to the physical machine (PNF). This metric is used to directly quantify and analyze the delay jitter impact introduced by the intermediate Ethernet switch.

Note: RTT stands for Round-Trip Time. To verify the stability of the split architecture under non-ideal fronthaul conditions, this study simulates network delays using Linux Traffic Control (TC) on the uplink ($T_5 - T_4$) and downlink ($T_2 - T_1$) segments.

To validate the proposed performance optimization in a realistic network environment, the system design utilizes a commercial-grade deployment framework:

- **O-RAN Split 7.2x Interoperability.**

Although the Small Cell Forum (SCF) has standardized the **network**

Functional Application Platform Interface (nFAPI) [?] for the MAC-PHY split (Option 6), commercial Radio Units (O-RUs) natively support O-RAN Split Option 7.2x. To achieve seamless integration with open-source protocol stacks like OpenAirInterface [?], this system bridges these interfaces to ensure full-scale commercial interoperability.

- **Hybrid System Architecture for End-to-End Verification.**

To verify the performance of nFAPI in a fully commercial ecosystem, this paper adopts a **hybrid architecture** (Split Option 6 + Split Option 7.2). In this model, the system provides a Translation/Interworking Layer that enables the nFAPI-driven MAC/High-PHY to successfully interact with the commercial Pegatron O-RU. This setup allows the use of **Commercial Off-The-Shelf (COTS)** user equipment to verify nFAPI performance in a real-world end-to-end environment.